PART I:

1. Create a Java program using bitwise operators that will produce the output: true, 10, false, 15.

```java
package bsit_activities;

public class BitwiseOperators {

    public static void main(String[] args) {

        // conditional bitwise AND
        boolean a = (2>1) & (0<9);
        System.out.println(a);   //true

        // computational bitwise AND
        int b = 14 & 11;
        System.out.println(b);   //10

        // conditional bitwise OR
        boolean c = (6>7) | (0>9);
        System.out.println(c);   //false

        // computational bitwise OR
        int d = 12 | 3;
        System.out.println(d);   //15
    }

}
```

```
33\bin' 'bsit_
ors'
true
10
false
15
PS C:\Users\fr
SCode\Java-Pro
```

2. Create a Java program using the ternary operator that will produce the output: true, false, 25, 30.

```java
package bsit_activities;

public class TernaryOperator {
    public static void main(String[] args) {

        int a = 67, b = 78;

        boolean c = (a<b)? true:false;
        System.out.println(c);   //true

        boolean d = (a>b)? true:false;
        System.out.println(d);   //false

        int e = (a<b)? 25:30;
        System.out.println(e);   //25

        int f = (a>b)? 25:30;
        System.out.println(f);   //30

    }
}
```

```
jdt_ws\Java-
it_activitie
true
false
25
30
PS C:\Users\
SCode\Java-P
```

3. Create a Java program using assignment operators that will produce the output: Addition: 120, Subtraction: 70, Multiplication: 144, Division: 16.

Operators.java | BitwiseOperators.java | StudentObjects.java | AssignmentOperators.java 2

bsit_activities > AssignmentOperators.java > Java > AssignmentOperators > main(String[] args)

```java
package bsit_activities;

public class AssignmentOperators {
    Run | Debug | Run main | Debug main
    public static void main(String[] args) {

        int num;

        // Addition
        num = 100;
        num += 20;
        System.out.println("Addition: " + num);

        // Subtraction
        num = 100;
        num -= 30;
        System.out.println("Subtraction: " + num);

        // Multiplication
        num = 12;
        num *= 12;
        System.out.println("Multiplication: " + num);

        // Division
        num = 64;
        num /= 4;
        System.out.println("Division: " + num);
    }
}
```

```
edhat.java\jdt_ws\Java
33\bin' 'bsit_activit
rators'
Addition: 120
Subtraction: 70
Multiplication: 144
Division: 16
PS C:\Users\franc\OneD
SCode\Java-Programs>
```

PART II:

sit_activities > UltraOperators.java > Java > UltraOperators > main(String[] args)

```java
package bsit_activities;

public class UltraOperators {
    Run | Debug | Run main | Debug main
    public static void main(String[] args) {

        //num  (arithmetic + assignment)
        int alpha = 30;
        alpha *= 3;                    //90
        System.out.println(alpha);    //1

        //bool (relational + logical)
        int beta = 15;
        boolean gamma = (beta > 10) && (beta != 20);
        System.out.println(gamma);    //2

        //num (bitwise shift)
        int delta = 4;
        int epsilon = delta << 2;     //4 * 2^2 = 16
        System.out.println(epsilon); //3

        //bool (logical + unary)
        int zeta = 8;
        boolean eta = !(zeta < 5) || (zeta == 8);
        System.out.println(eta);      //4

        //num (tricky increment)
        int theta = 5;
        int iota = theta++ + ++theta;
        // theta++ = 6, ++theta = 7, 5 + 7 = 12
        System.out.println(iota);     //5

        //num (modulus + assignment)
        int kappa = 50;
        kappa %= 9;                    //50 % 9 = 5
        System.out.println(kappa);    //6
    }
}
```

```
age\cbe166516bbb49
edhat.java\jdt_ws\
33\bin' 'bsit_acti
s'
90
true
16
true
12
5
PS C:\Users\franc\
SCode\Java-Program
```